

Report 1:

Project Description & Problem Statement

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CASSIA: A Constraint-Based Intelligent Agent for Coordinating Solar, Waste-Heat, Cooling, and Water Sourcing in AI Data Centers

Project Description

CASSIA is a simulation-based intelligent agent designed to optimize the dual resource consumption electricity and water of high-density AI data centers by deciding how limited solar energy and recovered waste heat are allocated across three activities: running AI workloads, cooling, and producing alternative water. Its decision-making combines constraint satisfaction (CSP) and heuristic search (A*) for scheduling and energy allocation with a rule-based knowledge base for cooling and water-source selection, and it replans every hour as conditions change.

Problem Statement

How can an intelligent agent dynamically coordinate energy resources, cooling and water systems, and flexible AI workloads to minimize the overall resource footprint of AI data centers without compromising computational performance?

AI data centers consume substantial amounts of energy and water to support computational workloads and cooling requirements. These resources are often managed across separate systems, limiting opportunities for coordinated resource optimization and sustainable operation. The specific problem is how to allocate solar energy and recovered heat each hour among workloads, cooling, and MD/FO water production, so total grid and municipal water use drop, without violating job deadlines or thermal limits.

Project Objectives

1. **Reduce Non-Renewable Energy Consumption:** Formulate and deploy an CSP-based job and cooling scheduler that increases on-site solar energy self-consumption and lowers total facility cooling energy usage.
2. **Minimize Water Usage Effectiveness (WUE):** Develop a constraint satisfaction model governing evaporative cooling and waste heat recovery that reduces direct facility water consumption while ensuring zero thermal throttling events for high-priority computational jobs
3. **Optimize Waste Heat Recovery:** Implement an optimal heuristic search algorithm (A^*) for waste heat allocation that routes low-grade exhaust heat into auxiliary heating loops or thermal energy storage during high-workload operations.

Expected Outcome

(i) A documented simulator with solar, waste-heat, cooling, workload, and MD/FO water-production models.
(ii) The CASSIA planner built on CSP, A*, and a rule base.
(iii) An evaluation against the baseline on grid electricity, municipal water, deadline compliance, and runtime. The results will show whether classical AI techniques can jointly reduce electricity and municipal water use.